

Simplifying Expressions with Square Roots (Part II) Homework

Simplify

Answers

① $\sqrt{19} \cdot \sqrt{19}$

19

② $\sqrt{31} \cdot \sqrt{31}$

31

③ $\sqrt{21} \cdot \sqrt{21}$

21

④ $\sqrt{6} \cdot \sqrt{5}$

$\sqrt{30}$

⑤ $\sqrt{3} \cdot \sqrt{7}$

$\sqrt{21}$

⑥ $\sqrt{11} \cdot \sqrt{2}$

$\sqrt{22}$

⑦ $\sqrt{7} \cdot \sqrt{5}$

$\sqrt{35}$

⑧ $\sqrt{2} \cdot \sqrt{10}$

$2\sqrt{5}$

⑨ $\sqrt{5} \cdot \sqrt{15}$

$5\sqrt{3}$

⑩ $\sqrt{6} \cdot \sqrt{10}$

$2\sqrt{15}$

⑪ $\sqrt{3} \cdot \sqrt{15}$

$3\sqrt{5}$

⑫ $\sqrt{14} \cdot \sqrt{2}$

$2\sqrt{7}$

⑬ $\sqrt{22} \cdot \sqrt{2}$

$2\sqrt{11}$

⑭ $3(4\sqrt{5})$

$12\sqrt{5}$

⑮ $7\sqrt{10}(-2)$

$-14\sqrt{10}$

⑯ $-8(-10\sqrt{6})$

$80\sqrt{6}$

⑰ $3\sqrt{5}(-8)$

$-24\sqrt{5}$

⑱ $2\sqrt{7}(4\sqrt{2})$

$8\sqrt{14}$

⑲ $-5\sqrt{3}(-8\sqrt{5})$

$40\sqrt{15}$

⑳ $(2\sqrt{10})(5\sqrt{7})$

$10\sqrt{70}$

㉑ $(-8\sqrt{2})(3\sqrt{3})$

$-24\sqrt{6}$

㉒ $(-4\sqrt{11})(-2\sqrt{6})$

$8\sqrt{66}$

㉓ $7\sqrt{10}(\sqrt{3})$

$7\sqrt{30}$

㉔ $\sqrt{5}(2\sqrt{11})$

$2\sqrt{55}$

㉕ $-3\sqrt{2}(-5\sqrt{6})$

$30\sqrt{3}$

㉖ $2\sqrt{8} \cdot 3\sqrt{3}$

$12\sqrt{6}$

㉗ $5\sqrt{3}(-8\sqrt{6})$

$-120\sqrt{2}$

㉘ $-3\sqrt{5}(2\sqrt{10})$

$-30\sqrt{2}$

㉙ $(-4\sqrt{8})(-3\sqrt{5})$

$24\sqrt{10}$

㉚ $2\sqrt{2}(\sqrt{8})$

$2\sqrt{16} = 2 \cdot 4 = 8$

㉛ $-\sqrt{18}(6\sqrt{2})$

$-6\sqrt{36} = -6 \cdot 6 = -36$

㉜ $3\sqrt{2}(4 + 2\sqrt{6})$

$12\sqrt{2} + 12\sqrt{3}$

㉝ $-5\sqrt{3}(\sqrt{10} - 1)$

$-5\sqrt{30} + 5\sqrt{3}$

$$(34) 4\sqrt{2}(8\sqrt{7}-3+6\sqrt{10})$$

$$32\sqrt{14}-12\sqrt{2}+48\sqrt{5}$$

$$(35) 6\sqrt{5}(-\sqrt{3}+2\sqrt{8}+10)$$

$$-6\sqrt{15}+24\sqrt{10}+60\sqrt{5}$$

$$(36) \frac{\sqrt{40}}{\sqrt{8}}$$

$$\sqrt{5}$$

$$(37) \frac{\sqrt{27}}{\sqrt{3}}$$

$$3$$

$$(38) \frac{\sqrt{48}}{\sqrt{6}}$$

$$2\sqrt{2}$$

$$(39) \frac{\sqrt{50}}{\sqrt{5}}$$

$$\sqrt{10}$$

Rationalize the denominators of the following fractions.

$$(40) \frac{8}{\sqrt{3}}$$

$$\frac{8\sqrt{3}}{3}$$

$$(41) \frac{10}{\sqrt{5}}$$

$$2\sqrt{5}$$

$$(42) -\frac{2}{\sqrt{6}}$$

$$-\frac{\sqrt{6}}{3}$$

$$(43) \frac{1}{\sqrt{8}}$$

$$\frac{\sqrt{2}}{4}$$

$$(44) \frac{3}{\sqrt{20}}$$

$$\frac{3\sqrt{5}}{10}$$

$$(45) \frac{-5}{\sqrt{18}}$$

$$\frac{-5\sqrt{2}}{6}$$

$$(46) \frac{7}{\sqrt{90}}$$

$$\frac{7\sqrt{10}}{30}$$

$$(47) \sqrt{\frac{3}{16}}$$

$$\frac{\sqrt{3}}{4}$$

$$(48) \sqrt{\frac{11}{64}}$$

$$\frac{\sqrt{11}}{8}$$

$$(49) \sqrt{\frac{17}{100}}$$

$$\frac{\sqrt{17}}{10}$$

$$(50) \sqrt{\frac{25}{6}}$$

$$\frac{5\sqrt{6}}{6}$$

$$(51) \sqrt{\frac{49}{10}}$$

$$\frac{7\sqrt{10}}{10}$$

$$(52) \sqrt{\frac{4}{7}}$$

$$\frac{2\sqrt{7}}{7}$$

$$(53) \sqrt{\frac{3}{5}}$$

$$\frac{\sqrt{15}}{5}$$

$$(54) \sqrt{\frac{10}{13}}$$

$$\frac{\sqrt{130}}{13}$$

$$(55) \sqrt{\frac{27}{10}}$$

$$\frac{3\sqrt{30}}{10}$$

$$(56) \sqrt{\frac{50}{7}}$$

$$\frac{5\sqrt{14}}{7}$$

$$(57) \sqrt{\frac{28}{5}}$$

$$\frac{2\sqrt{35}}{5}$$

$$(58) \sqrt{\frac{3}{20}}$$

$$\frac{\sqrt{15}}{10}$$

$$(59) \sqrt{\frac{7}{32}}$$

$$\frac{\sqrt{14}}{8}$$

$$(60) \sqrt{\frac{1}{72}}$$

$$\frac{\sqrt{2}}{12}$$

$$(61) \sqrt{\frac{45}{8}}$$

$$\frac{3\sqrt{10}}{4}$$

$$(62) \sqrt{\frac{20}{63}}$$

$$\frac{2\sqrt{35}}{21}$$

$$(63) \sqrt{\frac{245}{48}}$$

$$\frac{7\sqrt{15}}{12}$$

* You may
use a
calculator.